SPA617 Radio Astronomy Laboratory
Antenna Characterization

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1 Aim

To observe the received power at different distances and to measure the co-polarization and cross-polarization beam patterns of the given antenna.

2 Objectives

- To check how the received power changes with distance.
- To record E-plane and H-plane beam patterns for co-polarization and cross-polarization.
- To calculate the half-power beamwidth (HPBW) from the co-polarized patterns.
- To estimate antenna directivity from the measured beamwidths.

3 Theory

An antenna is a transducer. While transmitting, it converts electrical power into electromagnetic waves. While receiving, it converts the incident electromagnetic waves back into an electrical signal. In this experiment, one antenna is used as the transmitter and the other one is used as the receiver.

A dipole has two equal metal arms and is normally fed at the centre. Its total length is approximately $\lambda/2$. A monopole is like half of a dipole placed above a conducting ground plane, and its length is approximately $\lambda/4$. The ground plane acts like the other half of the dipole. The antenna does not radiate or receive equally in all directions, and this directional behaviour is what is measured as a beam pattern.

3.1 Received power and distance

For free-space propagation, the received power reduces when the separation R between the antennas is increased. Keeping the frequency and the antenna alignment fixed,

$$P_r \propto \frac{1}{R^2}. \tag{1}$$

The received power can also be written using the Friis transmission equation:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2, \tag{2}$$

where P_t is the transmitted power, G_t and G_r are the transmitter and receiver gains, and λ is the wavelength. In the lab, reflections and small alignment errors make the reading deviate from this ideal relation. In dB form, this gives

$$P_r(\text{dBm}) = C - 20\log_{10} R, \tag{3}$$

where C is a constant for one setup. Therefore, a plot of P_r versus $20\log_{10} R$ should ideally have slope close to -1. Also, if the distance is doubled, the expected reduction in received power is

$$20\log_{10}(2) = 6.02 \text{ dB}. \tag{4}$$

3.2 Beam pattern and polarization

The beam pattern shows how the received power changes when the receiving antenna is rotated. The direction with highest received power is the main lobe. The lobe opposite to it is called the back lobe. Smaller side lobes can occur because of the antenna pattern, reflections, and measurement error.

The E-plane contains the electric field direction and the direction of propagation. The H-plane is perpendicular to the E-plane. In the co-polarization case, the transmitting and receiving antennas have the same electric-field orientation, so the received power is expected to be high. In the cross-polarization case, the electric fields are perpendicular, so the received power is expected to be lower.

Since received power in dBm is negative, it cannot be used directly as a polar-plot radius. For plotting, the readings are converted to normalized linear power using

$$P_{\text{norm}} = 10^{(P_{\text{out}} - P_{\max})/10}. \tag{5}$$

Here, $P_{\max}$ is the largest measured power for that individual scan. The peak of each plotted pattern is then one.

3.3 Half-power beamwidth and directivity

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where received power is 3 dB lower than the peak value. A smaller HPBW means a more focused beam.

Using the E-plane and H-plane HPBWs in radians, the approximate directivity is

$$D \approx \frac{4\pi}{\theta_E \theta_H}. \tag{6}$$

This is an approximation, but it gives a useful estimate of how focused the antenna beam is.

4 Components Required

- RF signal generator and spectrum analyser

- Transmitting and receiving antennas with stands
- RF cables, adapters, and 50 Ω termination
- Measuring tape and angular scale

5 Experimental Setup and Method

The transmitting antenna was connected to the signal generator and the receiving antenna was connected to the spectrum analyser. The antenna heights and source power were kept same for one set of readings. For the distance measurement, the separation was changed while the antennas were kept aligned. For the beam patterns, the receiver was rotated in steps of 20° and the received power was noted.

For co-polarization, the electric fields of both antennas were kept in the same direction. For cross-polarization, one antenna was rotated by 90°. The angular scans in the notebook are marked around 1.8 GHz with $P_{\text{in}} = 0\,\text{dBm}$. The notebook does not write the plane name on two full-circle scans; they are used below as co-polarized H-plane and cross-polarized E-plane based on their order with the clearly labelled co-E and cross-H scans. This label can be changed easily if the lab record uses a different order.

6 Observations and Graphs

6.1 Effect of distance

The clean distance set in the notebook is marked near 1.34 GHz with input power 0 dBm. The first distance set above it contains overwritten values, so it is not used for a numerical fit. It is better not to force a calculation from a value which is not readable.

Table 1: Received power versus antenna separation.

Distance R (m)	$20\log_{10} R$ (dB)	P_r (dBm)
8.12	18.191	-61.2
7.85	17.897	-54.7
6.58	16.365	-51.2

The fitted relation for these three readings is

$$P_r = -4.425(20\log_{10} R) + 21.660, \qquad R^2 = 0.731. \tag{7}$$

Ideally, free-space propagation gives a slope close to -1 in this plot and a 6.02 dB reduction when distance doubles. The fitted data would give 26.64 dB for doubling, which is clearly not reliable with only three readings. The likely reasons are alignment change, reflections from the room, and the small number of points. The useful observation is still that the received power becomes lower at larger separation.

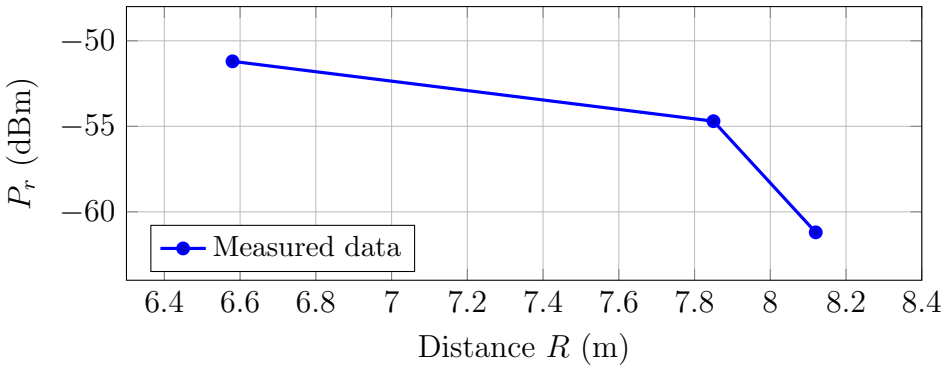


Figure 1: Received power versus distance. The line only joins the measured points.

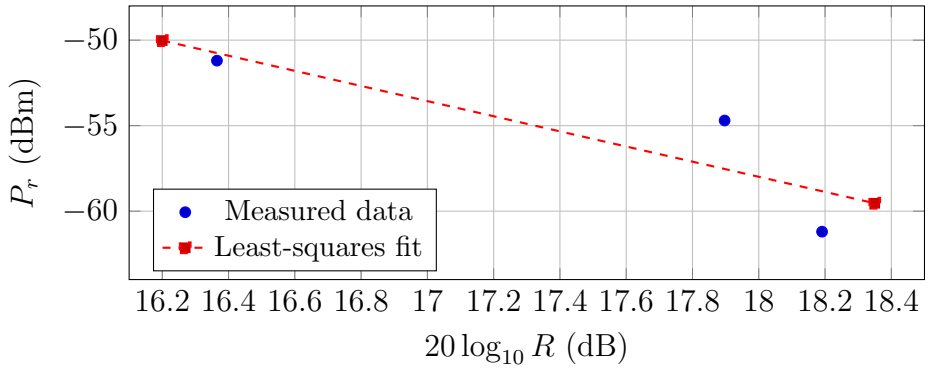


Figure 2: Received power versus $20 \log_{10} R$.

6.2 Beam-pattern readings

All values below are the received power P_{out} in dBm. The tables retain the readings from the notebook, including small differences between 0° and 360°.

The supplied polar plots use normalized linear power, $P_{\text{norm}} = 10^{(P_{\text{out}} - P_{\max})/10}$, as the radius. Each plot is placed directly after the table from which it was made.

Table 2: Co-polarized E-plane readings.

θ (°)	P_{out} (dBm)	θ (°)	P_{out} (dBm)
0	-53.0	200	-60.1
20	-54.1	220	-70.3
40	-55.3	240	-58.1
60	-56.4	260	-55.1
80	-61.2	280	-56.1
100	-64.1	300	-57.6
120	-61.02	320	-62.4
140	-70.4	340	-57.4
160	-68.8	360	-53.4
180	-61.7		

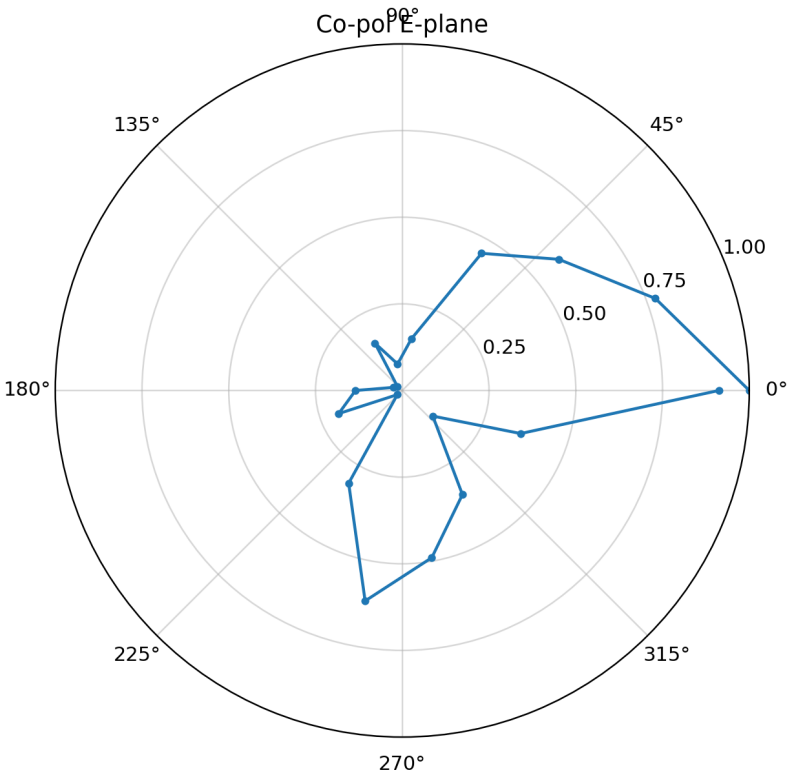


Figure 3: Normalized co-polarized E-plane beam pattern from Table 2.

Table 3: Co-polarized H-plane readings (plane label inferred from notebook sequence).

ϕ (°)	P_{out} (dBm)	ϕ (°)	P_{out} (dBm)
0	-58.8	200	-64.5
20	-60.3	220	-70.5
40	-66.9	240	-74.7
60	-75.0	260	-62.6
80	-72.0	280	-61.5
100	-70.0	300	-60.8
120	-73.9	320	-68.3
140	-66.5	340	-62.3
160	-65.2	360	-58.8
180	-67.13		

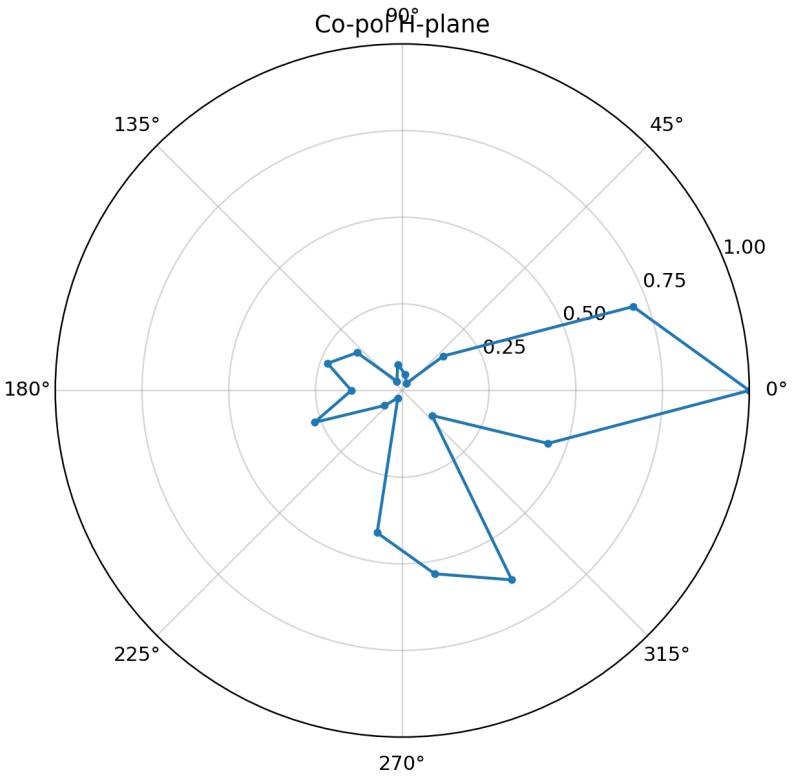


Figure 4: Normalized co-polarized H-plane beam pattern from Table 3.

6.3 Cross-polarization beam patterns

The receiver was rotated through the same angular range after making the transmitting and receiving electric fields perpendicular. The cross-polarized E-plane and H-plane readings are plotted separately below.

Table 4: Cross-polarized E-plane readings (plane label inferred from notebook sequence).

θ (°)	P_{out} (dBm)	θ (°)	P_{out} (dBm)
0	-64.71	200	-81.37
20	-63.79	220	-77.46
40	-64.67	240	-75.11
60	-65.10	260	-70.75
80	-71.34	280	-69.52
100	-73.50	300	-68.67
120	-74.68	320	-64.21
140	-80.47	340	-63.65
160	-75.21	360	-64.16
180	-77.48		

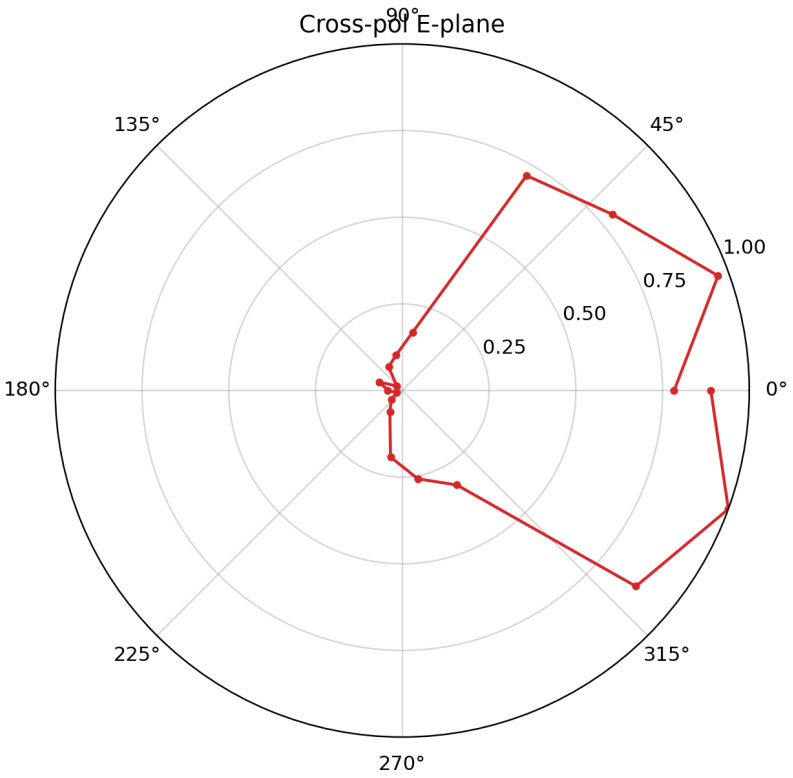


Figure 5: Normalized cross-polarized E-plane beam pattern from Table 4.

Table 5: Cross-polarized H-plane readings.

ϕ (°)	P_{out} (dBm)	ϕ (°)	P_{out} (dBm)
0	-65.5	200	-69.8
20	-76.5	220	-69.2
40	-81.5	240	-72.9
60	-70.6	260	-64.2
80	-73.9	280	-68.7
100	-74.0	300	-61.1
120	-65.0	320	-59.2
140	-73.2	340	-60.2
160	-70.5	360	-63.7
180	-70.25		

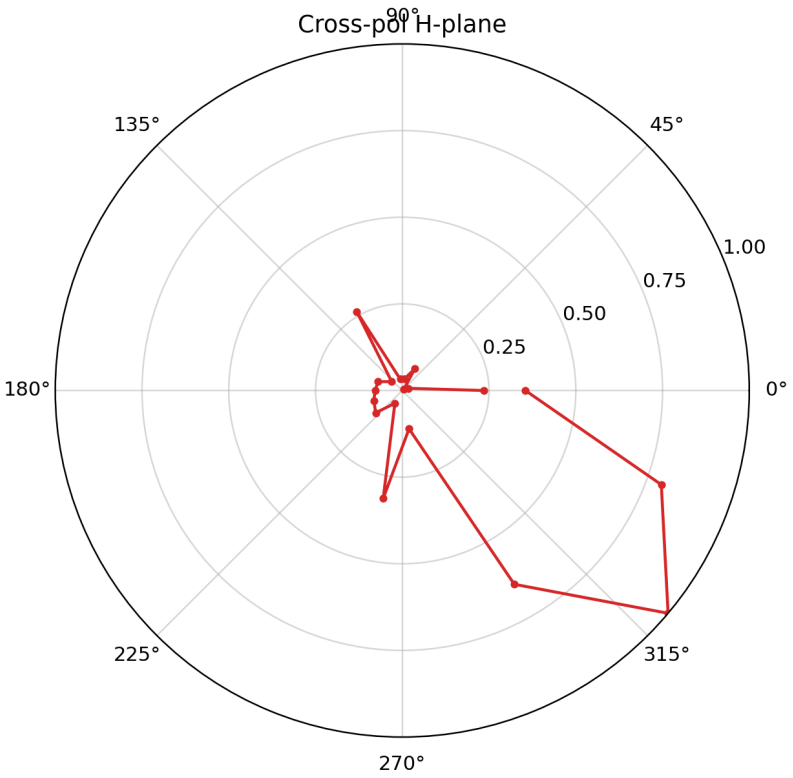


Figure 6: Normalized cross-polarized H-plane beam pattern from Table 5.

Figures 5 and 6 are the required E-plane and H-plane beam patterns for the cross-polarization configuration. Compared with the co-polarized plots, the cross-E plot has a lower maximum received power. The cross-H plot shows a peak near 320° instead of the reference direction, which can be due to small antenna misalignment and room reflections.

7 Half-Power Beamwidth and Directivity

The HPBW is found at the points which are 3 dB below the maximum co-polarized received power. Linear interpolation was used between the nearest 20° readings.

7.1 E-plane HPBW

For the E-plane, the maximum measured power is

$$P_{\max,E} = -53.0\,\text{dBm}. \tag{8}$$

So the half-power level is

$$P_{-3\text{dB},E} = P_{\max,E} - 3 = -56.0\,\text{dBm}. \tag{9}$$

On the positive-angle side, the crossing is between 40° and 60°:

$$\theta_{E,+} = 40^\circ + \frac{-56.0 - (-55.3)}{-56.4 - (-55.3)}(20^\circ) \tag{10}$$

$$= 52.73^\circ. \tag{11}$$

Across the 0°/360° boundary, the other crossing is between 340° and 360°:

$$\theta_{E,-} = 340^\circ + \frac{-56.0 - (-57.4)}{-53.4 - (-57.4)}(20^\circ) \tag{12}$$

$$= 347.00^\circ = -13.00^\circ. \tag{13}$$

Therefore,

$$\boxed{\text{HPBW}_E = 52.73^\circ - (-13.00^\circ) = 65.73^\circ = 1.147\text{ rad.}} \tag{14}$$

7.2 H-plane HPBW

For the H-plane,

$$P_{\max,H} = -58.8\,\text{dBm}, \qquad P_{-3\text{dB},H} = -61.8\,\text{dBm}. \tag{15}$$

The positive-angle crossing is between 20° and 40°:

$$\phi_{H,+} = 20^\circ + \frac{-61.8 - (-60.3)}{-66.9 - (-60.3)}(20^\circ) \tag{16}$$

$$= 24.55^\circ. \tag{17}$$

The other crossing is between 340° and 360°:

$$\phi_{H,-} = 340^\circ + \frac{-61.8 - (-62.3)}{-58.8 - (-62.3)}(20^\circ) \tag{18}$$

$$= 342.86^\circ = -17.14^\circ. \tag{19}$$

Hence,

$$\boxed{\text{HPBW}_H = 24.55^\circ - (-17.14^\circ) = 41.69^\circ = 0.728\text{ rad.}} \tag{20}$$

Using the beamwidth approximation,

$$D \approx \frac{4\pi}{\theta_E \theta_H} \tag{21}$$

$$= \frac{4\pi}{(1.147)(0.728)} \tag{22}$$

$$= 15.06. \tag{23}$$

Therefore the estimated directivity is

$$\boxed{D \approx 15.06 \quad \text{or} \quad 10\log_{10}(D) = 11.78 \text{ dBi.}} \tag{24}$$

8 Results

Table 6: Important measured and calculated results.

Quantity	Symbol	Value
Co-polarized E-plane peak power	$P_{\max,E}$	-53.0 dBm
Co-polarized H-plane peak power	$P_{\max,H}$	-58.8 dBm
E-plane HPBW	θ_E	65.73° (1.147 rad)
H-plane HPBW	θ_H	41.69° (0.728 rad)
Estimated directivity	D	15.06 (11.78 dBi)

The cross-E peak is about -63.65 dBm, which is around 10.65 dB below the co-E peak. The cross-H peak is close to the co-H peak in this data. This can happen due to imperfect 90° alignment, reflections from nearby objects, or rotation reference error. So the cross-polar plots should be treated mainly as the measured pattern shape, not as a precise isolation measurement.

9 Conclusion

The received power generally reduced when the antenna separation was increased. The clean three-point distance data has noticeable scatter, so it is not enough to claim an accurate free-space loss value.

The co-polarized beam patterns gave a main lobe near the reference direction. From the -3 dB points, the E-plane HPBW was found to be 65.73° and the H-plane HPBW was 41.69°. Using these values, the approximate directivity was 15.06, or 11.78 dBi.

The cross-polarized readings were lower than the co-polarized E-plane readings, as expected. Some asymmetry and unexpected peaks are present in the scans. This can be because the antenna was not perfectly aligned, the measurement was done in a room with reflections, and the angle was changed in 20° steps.

Reference

SPA617 Radio Astronomy Laboratory, *Antenna Characterization*, Revision 2.0, August 2026.

AI Usage

ChatGPT was used for, plotting the data, and formatting this report in LaTeX.